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Visualising Space and Time

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SET09123 – Interactive Data Visualisation

Lecture Summary

- Spatial Data
- Geographical Visualisations
- Temporal Data
- Time Series Visualisations

Spatial Data

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How would you define Spatial Data?

Spatial Data

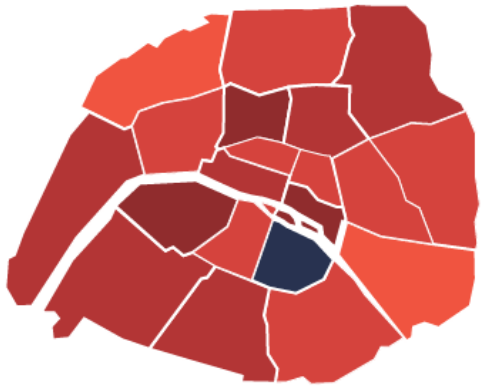
- Spatial Data is data where one or more attributes hold a spatial semantic
- Numerical: latitude and longitude, coordinates on a grid
- Categorical: names of cities, regions or countries

Encoding Spatial Data

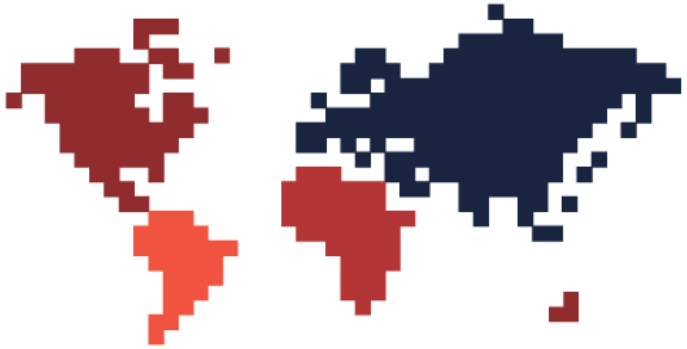
- In the large majority of cases, the position visual channels are automatically used to encode these spatial attributes
  - Directly positioning with coordinates
  - Positioning with distances between objects

- Other channels are then used to encode further attributes

Space – Encoding Regions



Choropleth Map



Cartogram

Isoline Map

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Space – Encoding Landmarks

Pin Map

Route Map

Dot Density Map

Image Source: datavizproject.com

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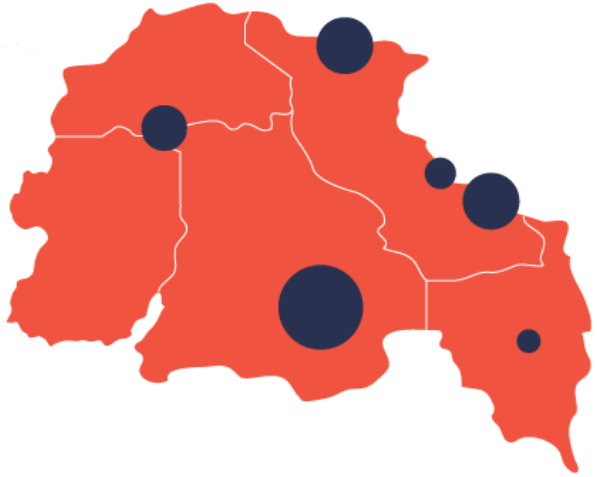
Space – Superimposition

Flow Map



Bubble Map

Connection Map

Image Source: datavizproject.com

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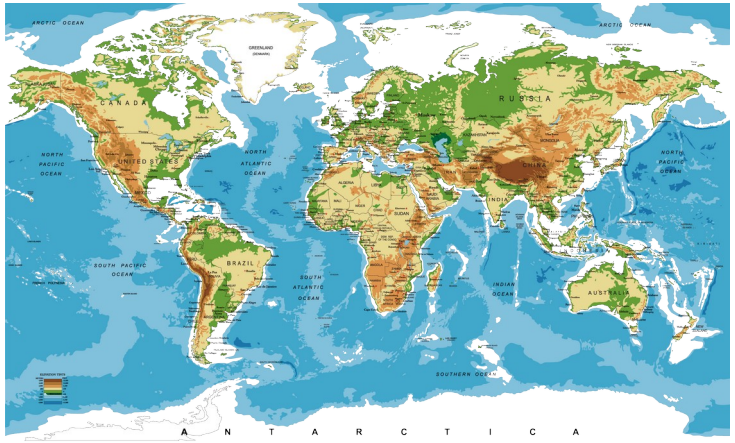
Geographical Data

- Geographical Information Systems (GIS) and their visualisation is a entire separate field in itself

- When it comes to visualising space, the key is to map* from a non-Euclidean *geographical coordinate system* onto a 2D *orthogonal coordinate system*



*pun intended

Geographical Projections

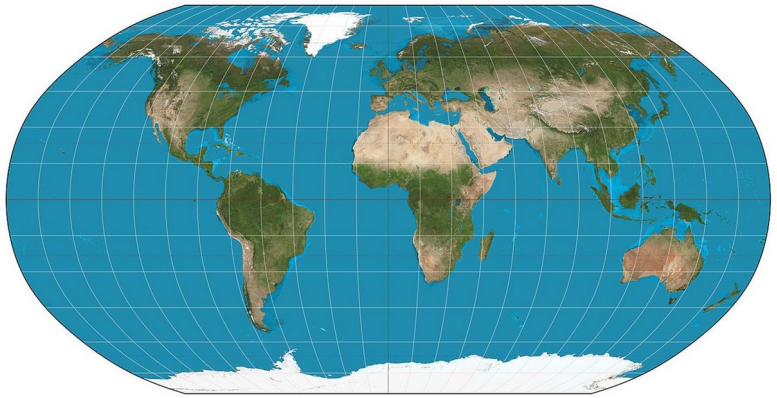
- Typically, one would then use a **projection** to visualise maps
  - A method for mapping points/lines/areas between different coordinate systems

- There are **many** projection types for geographical data

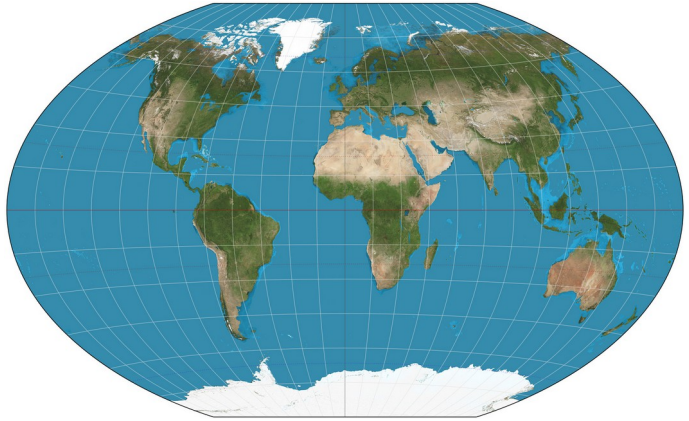
- Their aim is to minimise distortions:
  - Area
  - Distances
  - Directions

Geographical Projections

Mercator: all meridians are parallel, creates huge distortions at the poles

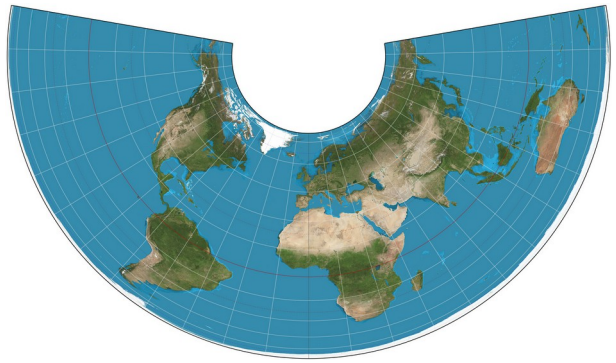


Robinson: reduces the pole distortions from Mercator

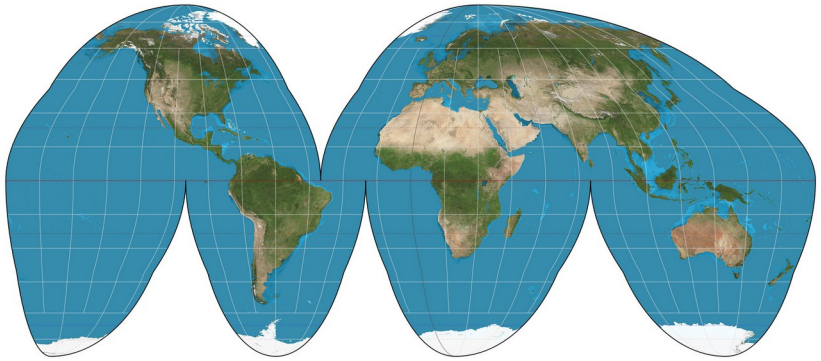


Winkel Tripel: minimises area, distances and direction distortions

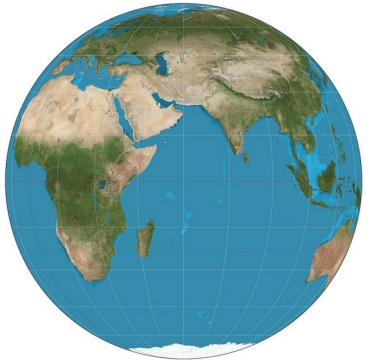
Geographical Projections



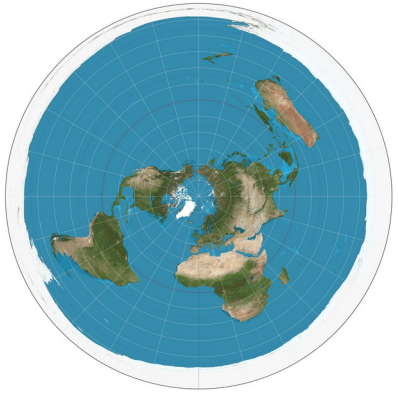
Albers conic: minimises area distortions



Goode homolosine: minimises land area distortions



Orthographic: Hemisphere projected onto a plane



Azimuthal equidistant: conserves distance from the centre point

Waterman butterfly: Project globe onto octahedron

Peirce's quincuncial: Can be tiled

Temporal Data

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How would you define Temporal Data?

Temporal Data

- Temporal Data is data where one or more attributes hold a temporal semantic
- Numerical: hours, minutes, days, years
- Categorical: week days, seasons, months, years

Time Series

- We use the term **Time Series** when, for the same real-world entity, there are several instances of the same attribute recorded at different times
- Distribution of events over a time period
- Evolution of several attributes over time

Representing Time Series

There are several strategies to encode time:

- Use a visual channel to encode time
- Encode time with time
- Embed time in other layouts

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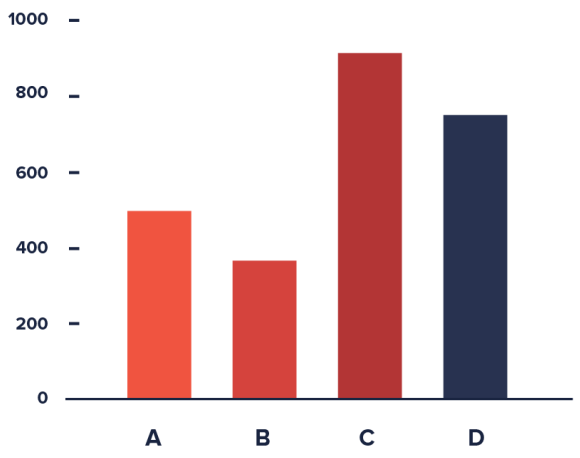
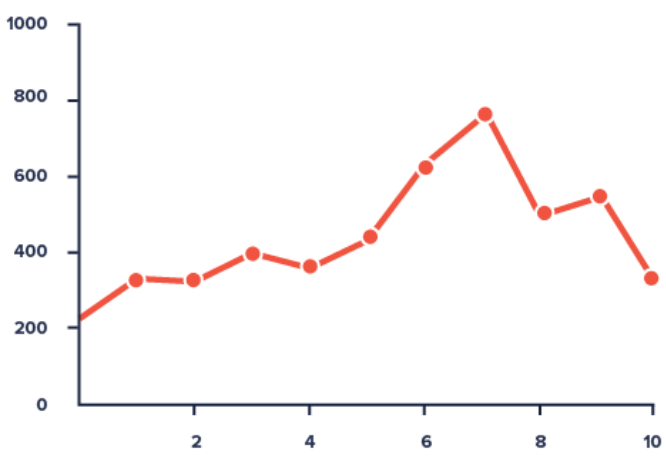
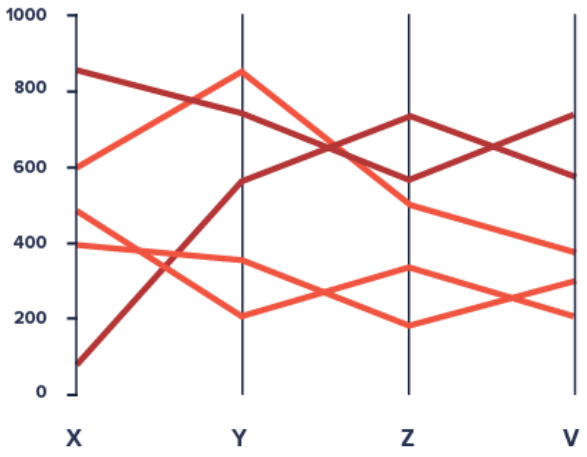
Time Axis



Bar Chart



Line Chart



Parallel Coordinates

Image Source: datavizproject.com

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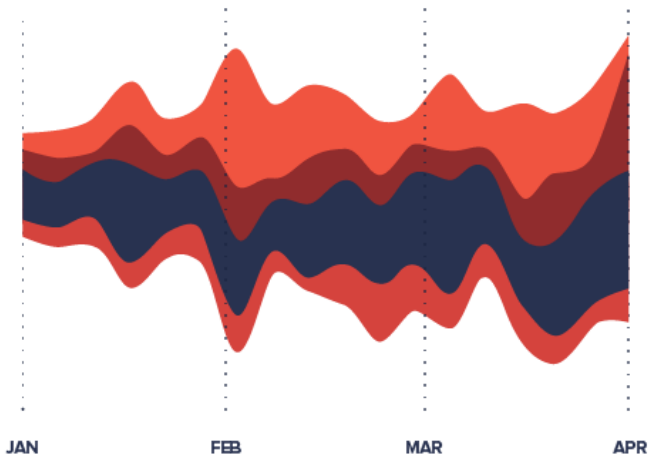
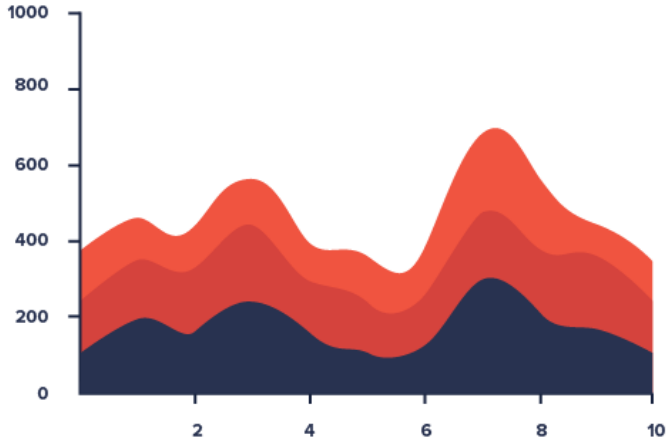
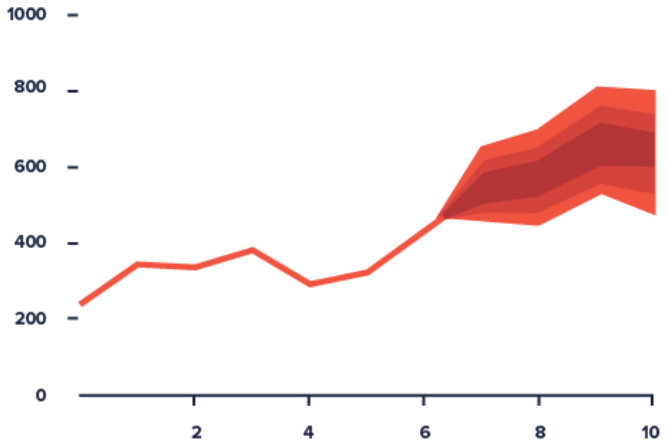
Areas



Stream Graph



Stacked Area Chart



Fan Chart

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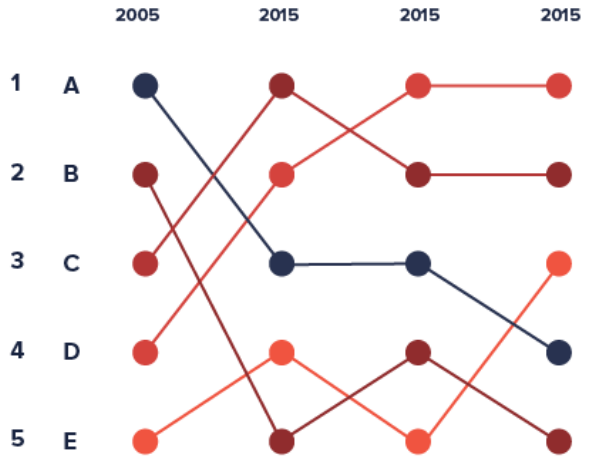
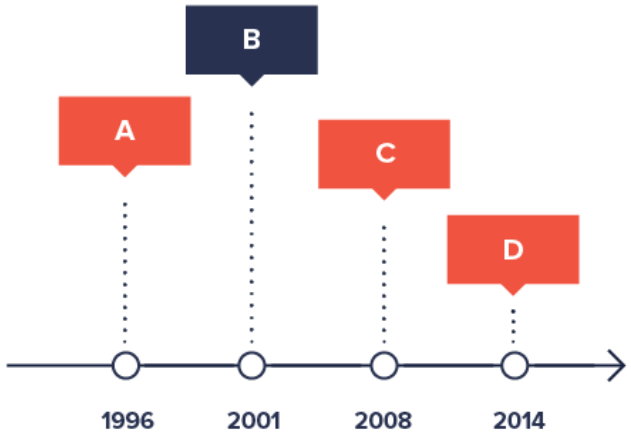
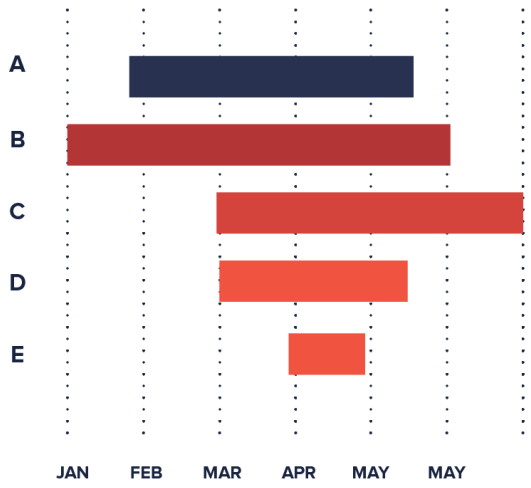
Time Specific Charts



Bump Chart



Timeline



Gantt Chart

Image Source: datavizproject.com

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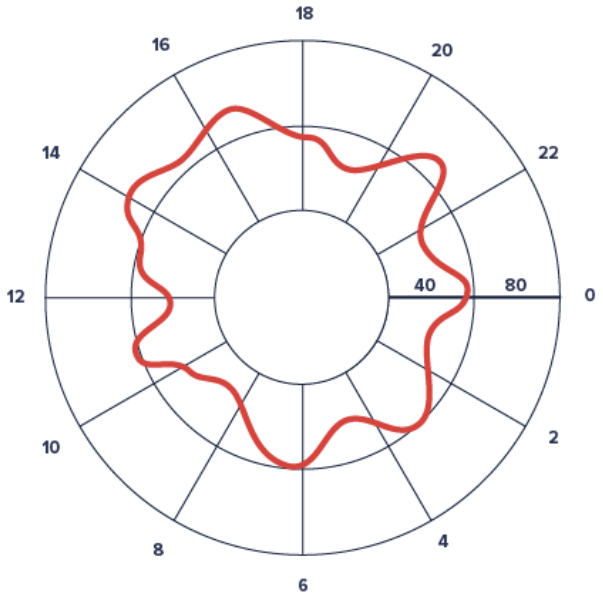
Cycles

Polar Area Chart

Radial Histogram



Radial Line Graph

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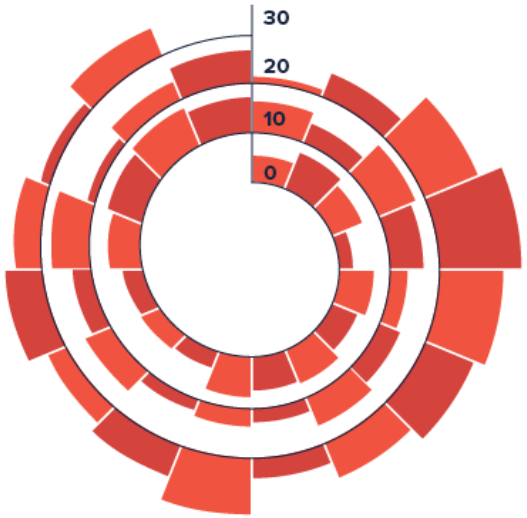
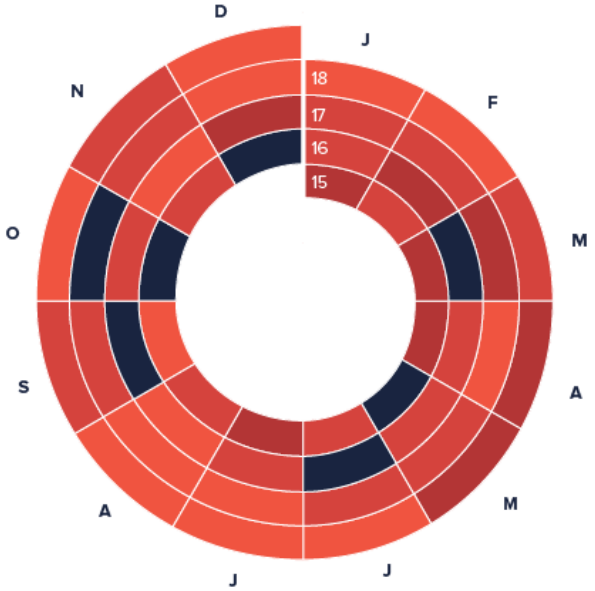
Ongoing Cycles



Spiral Histogram



Spiral Heat Map

Image Source: datavizproject.com

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Representing Time Series

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There are several strategies to encode time:

- Use a visual channel to encode time
- Encode time with time
- Embed time in other layouts

Encode Time with Time

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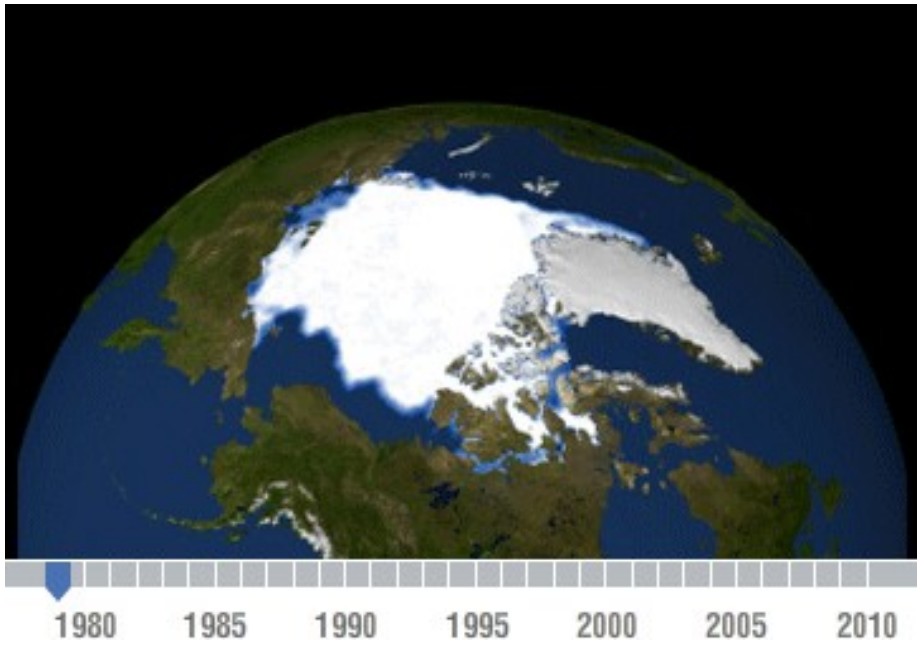
- The idea is to produce several **Timeslices**
- Each slice represent the state of the dataset for one specific moment in the time period we visualise
- We then enable *navigation* through these slices

Animating Timeslices

- Navigate with **animation**

- Humans are especially good at detecting changes
- But, comparing these changes can be mentally demanding
- In general, animation gives an *overview*

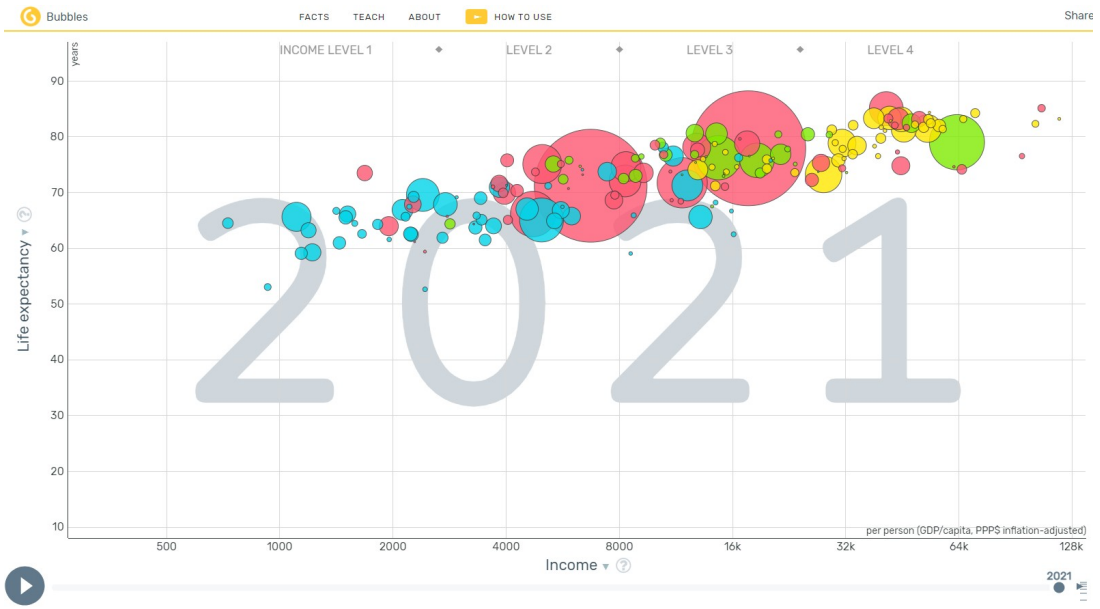


Source: wikimedia

Interacting with Timeslices

- Allowing users to navigate at will with **interaction** is even more powerful

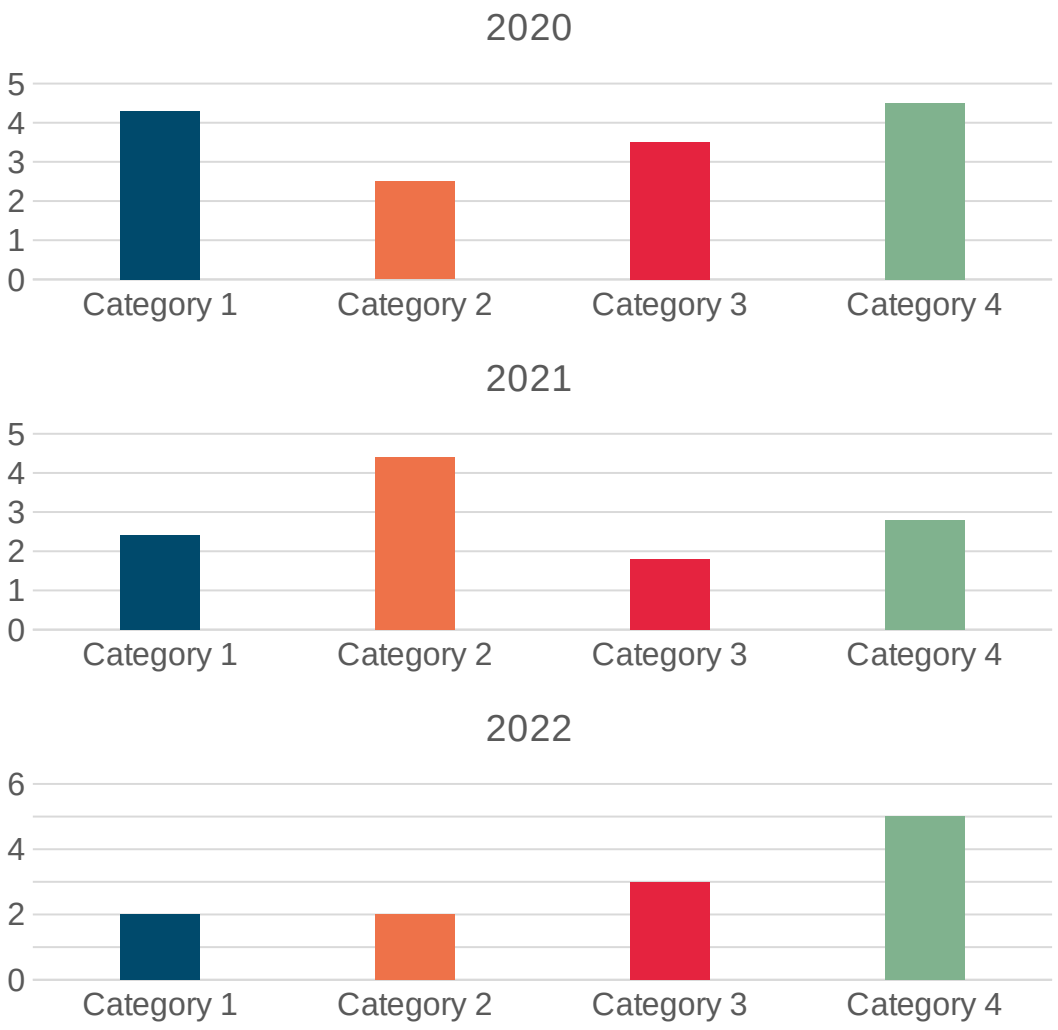
- And you can still include some animation



Example: [Gapminder]

Showing All Timeslices

- Animation and Interaction both rely on the users' short term memory to be effective
- **Small-multiples** are another way to present timeslices
- This is useful to look at thing in *details*



Representing Time Series

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There are several strategies to encode time:

- Use a visual channel to encode time
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Embed Time

- Typically used with spatial visualisations

- It can sometime lead to occlusion
- It’s not as easy for users to read

Conclusion

Key Points

- Spatial and Temporal data both relate to the **semantics** attached to one or more of the data attributes

- When showing spatial data, position visual channels are taken, hence we often rely on other channels to show more data
- Visualising geographical data involves the use of **projections**: 2D maps showing 3D features

- There are several ways to encode time on visualisation:
  - Make a **time axis**
  - Create and navigate **timeslices**
  - Embed time on other visualisations
- Polar displays are often well suited to show **cycles**